Pemphigus. S2 Guideline for Diagnosis and Treatment – Guided by the European Dermatology Forum (EDF) in Cooperation with the European Academy of Dermatology and Venereology (EADV)

Abstract

Background

Pemphigus encompasses a group of life-threatening autoimmune bullous diseases characterized by blisters and erosions of the mucous membranes and skin. Before the era of immunosuppressive treatment, the prognosis of pemphigus was almost fatal. Due to its rarity, only few prospective controlled therapeutic trials are available.

Objectives

A group of European dermatologists with expertise in basic and clinical pemphigus research aimed to define diagnostic and therapeutic guidelines for the management of patients with pemphigus.

Results

The group identified key statements of agreement or disagreement regarding the diagnosis and treatment of pemphigus. The revised final version of the guideline was submitted to the European Dermatology Forum (EDF) for final consensus

with the European Academy of Dermatology and Venereology (EADV) and the European Union of Medical Specialists (UEMS).

Introduction

Pemphigus is a group of life-threatening autoimmune blistering disorders characterized by flaccid blisters and erosions affecting the skin and mucous membranes. The severity of the disease stems from its progressive course, which can lead to increased catabolism, loss of body fluids and proteins, and secondary bacterial or viral infections—potentially resulting in sepsis and cardiac failure.

Prior to the availability of systemic corticosteroids, pemphigus was almost universally fatal within two years of diagnosis. The pathophysiology involves IgG autoantibodies targeting desmosomal adhesion proteins—specifically desmoglein 3 and/or desmoglein 1—on epidermal keratinocytes, leading to intraepithelial blister formation.

Pemphigus is rare, with an estimated incidence of approximately two new cases per 1 million people per year in Central Europe. The two main clinical variants are **pemphigus vulgaris (PV)** and **pemphigus foliaceus (PF)**.

The pathogenic role of anti-desmoglein 1/3 IgG antibodies is well established: injection of patient sera or affinity-purified IgG from pemphigus patients into neonatal mice reproduces the clinical and immunopathological features of the disease within 24 hours. In most patients, disease activity correlates closely with serum levels of desmoglein-reactive autoantibodies.

Due to the rarity of pemphigus, there are limited prospective, controlled clinical trials. Existing studies are often underpowered due to small patient numbers and lack statistically significant outcomes. Some trials have compared different prednisolone doses, intravenous corticosteroid pulses versus placebo, azathioprine versus mycophenolate mofetil, and the use of adjuvant therapies such as methotrexate, cyclosporine, cyclophosphamide, and high-dose intravenous immunoglobulins (IVIG).

The combination of systemic corticosteroids (prednisolone at 1.0–1.5 mg/kg/day) and corticosteroid-sparing immunosuppressive agents—most commonly azathioprine or mycophenolate mofetil—is widely considered first-line therapy.

Despite national efforts in countries like France and the United Kingdom, no internationally accepted treatment guidelines for pemphigus currently exist. To address this gap, a panel of European dermatologists with extensive experience in pemphigus research and clinical care has developed comprehensive diagnostic and therapeutic recommendations.

Methodology of Guideline Preparation

The working group of European dermatologists adopted a methodology previously used in the development of the French pemphigus guidelines to ensure a structured and evidence-based approach to formulating recommendations.